L1 · BRICK

GENERAL ARRANGEMENT & FEATURES

FIELD NODE — HANDHELD MESH BRICK

AF-L1-101

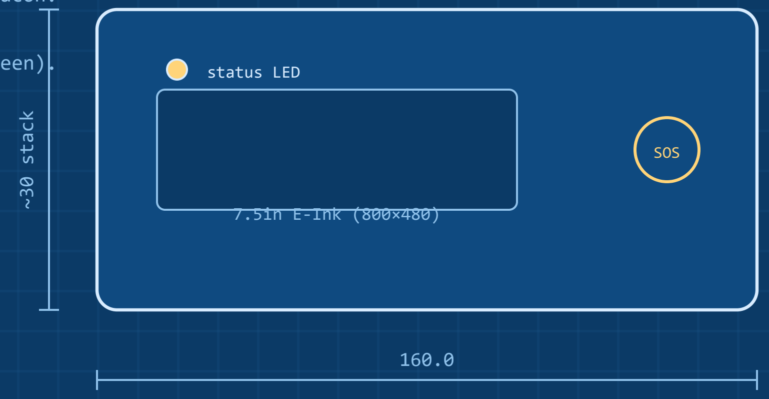
REV A · 2026-11-12

PROTOTYPE

WHAT THIS NODE IS

- One handheld brick that is simultaneously: mailbox, mesh repeater, public bulletin (E-Ink), and SOS beacon.
- Runs the ArcticChain agent (poc/arcticchain_agent.py) — HTTP API on :8080, SQLite store-and-forward.
- Every node holds an Ed25519 identity; envelopes signed + replay-protected (mesh-net crate, 46 tests green).
- Works with zero connectivity: LoRa mesh to neighbors, LTE backhaul only when a tower exists.

ENVELOPE — HANDHELD BRICK



shell: 160×80 mm face · ~32 mm thick w/ stack · ~100 g PETG

INTERNAL SYSTEM

CM5 + IO BOARD
compute · 4GB · headers

LTE HAT (SIM7600)
4G backhaul · GNSS · BT

E-INKE HAT 7.5in
public display · SPI0

LoRa SX1262
915 MHz mesh · SPI1

4×18650 TRAY
14.8 V → 5 V buck

STACK bottom→top: USB-C pwr → CM5 IO → LTE → E-Ink → LoRa

WHAT IT DOES FOR THE COMMUNITY

Function	How
Mailbox	Email/SMS-tier messages queue in SQLite and deliver when any path opens
Repeater	Forwards signed envelopes — every node extends coverage ~5–15 km
Bulletin	E-Ink shows alerts, notices, prices — readable in sunlight, zero power to display
SOS beacon	Front button pages responders over mesh when all else fails

ACCEPTANCE TEST (THE WHOLE PITCH IN ONE TEST)

- Unit A: POST /message {"to":"broadcast","text":"Mesh test"} → Unit B: GET /inbox returns it.
- Repeat with LTE in airplane mode — delivery with NO towers = the product.
- curl http://localhost:8080/health → {"lora":true,"lte":true,"uptime":...}

KNOWN GAPS (BY DESIGN — BUILD_GUIDE.md)

- LoRa send_frame/poll are TODOs — plaintext beacon until SX1262 driver + envelope land (wk 3).
- E-Ink is stub file-write; real Waveshare driver is Level-2.
- No power-path controller — battery swap reboots CM5 (acceptable PoC).

ARCTICCHAIN · RURAL NODE FLEET

PROPRIETARY & CONFIDENTIAL — EMEBET INC.

DRAWN: Surafel Negussie — Planner / Maker

SOLE PROPRIETARY OWNER
Emebet Inc. · emebetos.ca
registered Canadian company — Edmonton, AB

DWG AF-L1-101
REV A
SCALE NTS · SHEET 1/4

DESIGNED FOR RURAL & REMOTE CANADA — -45..+70 °C · IP65 · SOLAR-FIRST · COMMUNITY-OWNED

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MECHANICAL & ENVELOPE

PRINTED CHASSIS — PETG FIELD ENCLOSURE

AF-L1-102
REV A · 2026-11-12
PROTOTYPE

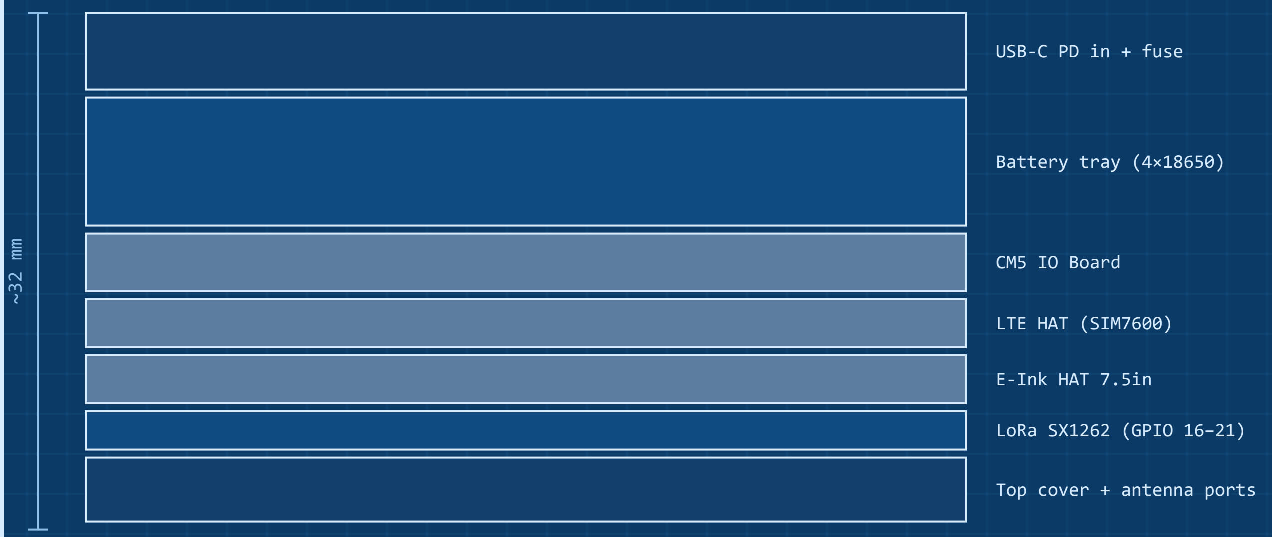
ENCLOSURE SPEC (FROM cad/level1_chassis.py → dimensions.json)

Parameter	Value	Parameter	Value
[illegible] (X)			
[illegible] (Y)			
[illegible] 32 mm cavity			
[illegible] (PETG) [illegible]nals			
[illegible]			
[illegible] (dear)			

PRINT SETTINGS

- Material: PETG (field-rated; NOT PLA — enclosure lives outside). REV-B target: PC or ASA for -45 °C.
- Infill 40%, walls 4, no supports required (verify in slicer preview).
- Files: cad/level1/chassis_shell.stl (540 tris) + battery_tray.stl (2012 tris).
- Post-print: deburr seams, dry-fit stack, conformal-coat rev for IP65 (silicone gasket at lid seam).

STACK-UP SECTION (SIDE VIEW)



ANTENNA KEEP-OUT (CRITICAL FOR RANGE)

- LTE whip: top edge, SMA bulkhead; ≥10 mm from LoRa antenna.
- LoRa 915 MHz: spring/whip at OPPOSITE end, vertical.
- NO metal (cells, buck inductor) within 15 mm of either antenna.
- Route through top grommets; seal with silicone.

MOUNTING & FIELD NOTES

- Handheld + pocket clip; SOS button reachable one-handed, glove-friendly rev-B.
- E-Ink readable in direct sun; updates only on new content (zero idle power).
- Operating target: -20..+50 °C gen-1; -45 °C requires LiFePO4/cold-rated cells rev-B.
- Two-person install NOT required — this is the personal node.

REVISION ROADMAP

Rev	Change	Driver	Status
[illegible]) NMC 18650, stub E-Ink			
[illegible] IP65 gasket, glove SOS			
[illegible] (BQ24075-class)			
[illegible] SX1262 envelope			

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ELECTRICAL — BOM, POWER & GPIO

CIRCUIT & PARTS SPECIFICATION

AF-L1-103

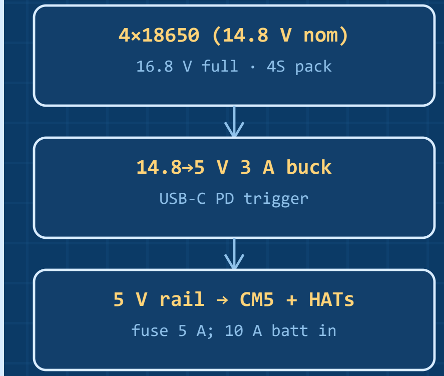
REV A · 2026-11-12

PROTOTYPE

BILL OF MATERIALS (PER UNIT — ORDER LIST ×2 FOR ACCEPTANCE TEST)

#	Part	Supplier class	Approx CAD	Qty
[illegible] IO Board				
[illegible] 800×480 SPI (Waveshare-class)				
[illegible] (GNSS+BT ok)				
[illegible] 915 MHz SPI (EBYTE E22-900M30S class)				
[illegible] pairs ×6				
[illegible] protected + 14.8V→5V 3A buck (USB-C PD)				
[illegible] (both shells + trays ~200 g)				

POWER TREE



4×18650 (14.8 V nom)
16.8 V full · 4S pack

14.8→5 V 3 A buck
USB-C PD trigger

5 V rail → CM5 + HATs
fuse 5 A; 10 A batt in

Runtime est: 4×3000 mAh ≈ 44 Wh →
LTE idle ~1–2 W, LoRa TX bursts ~0.1 W →
days–weeks autonomy; solar charge port rev-B.

GPIO ALLOCATION (NO CONFLICTS)

Signal	GPIO	Notes
[illegible] · 11 SCLK · 8 CE0		
[illegible]/refresh		
[illegible]		
[illegible]		
[illegible] 14/15)		
[illegible]		
[illegible] activity		

/boot/firmware/config.txt — REQUIRED OVERLAYS

```
dtparam=i2c_arm=on
dtparam=spi=on
dtparam=uart0=on
dtoverlay=spi1-3cs
dtoverlay=gpio-shutdown
disable_overscan=1
# E-Ink=spi0 (GPIO7-11) · LoRa=spi1 (GPIO16-21) — never share
```

STACK ORDER (BOTTOM → TOP)

- USB-C power + battery tray (bottom bay)
- CM5 IO Board — power, HDMI, USB, GPIO header
- 4G/LTE HAT — UART + USB AT commands
- E-Ink HAT — SPI0 + GPIO25 busy
- LoRa SX1262 — SPI1 GPIO16–21 (top)

WIRING RULES

- Standoffs M2.5 nylon at PCB corners; 2.5 mm headers between HATs.
- Thermal pad on CM5 SoC; keep buck inductor away from antennas.
- Double-check LoRa breakout silkscreen — CE/RESET/DIO numbering varies.

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REV A

SCALE NTS · SHEET 3/4

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SOFTWARE, ASSEMBLY & TEST

FIRMWARE + BUILD PROCEDURE

AF-L1-104
REV A · 2026-11-12
PROTOTYPE

SOFTWARE STACK

Component	Detail
OS	Raspberry Pi OS Lite 64-bit on 32 GB microSD; user 'arctic', SSH enabled
Agent	poc/arcticchain_agent.py — stdlib Python, systemd service arcticchain-agent.service
API :8080	GET /health · GET /inbox · GET /signal · POST /message {to,text}
Store	SQLite inbox/outbox — store-and-forward, survives days offline
Mesh-crypto	Mesh-net crate: Ed25519 sign · X25519 · ChaCha20-Poly1305 (Rust, 46 tests)
Roadmap	LoRa driver + envelope compat (wk3) → E-Ink driver (L2) → power-path (L2)

ASSEMBLY SEQUENCE (7 STEPS — ORDER_LIST.md)

① Print chassis_shell.stl + battery_tray.stl — PETG, 40% infill, 4 walls.

② Flash Pi OS Lite to both SD cards (Imager → SSH on, user 'arctic').

③ Bench-test bare CM5: boot, SSH, i2cdetect -y 1 — BEFORE stacking HATs.

④ Stack per WIRING.md; wire LoRa to GPIO 16–21 (spi1, never spi0).

⑤ Wire power: 4×18650 → 14.8→5 V buck → USB-C; fuse 5 A rail, 10 A input.

⑥ Deploy agent: copy arcticchain_agent.py + systemd unit (poc/SETUP.md §5).

⑦ Acceptance test: POST /message on A → GET /inbox on B → airplane-mode LTE → repeat.

FIELD ACCEPTANCE CRITERIA

- Two units exchange a mesh message with LTE disabled — verified on bench AND in field.
- Message latency < 60 s at ≤ 5 km line-of-sight (LoRa BW/SF tuned for range).
- Node survives -20 °C night + sun-loaded enclosure (+60 °C internal) — rev-B extends to -45 °C.
- Battery swap → clean reboot → inbox intact (SQLite on SD survives power loss).

FILES IN REPO

- cad/level1_chassis.py — parametric generator (CadQuery)
- cad/level1/*.stl + *.step + dimensions.json — validated
- poc/arcticchain_agent.py — hardware-bridge agent (stdlib only)
- poc/arcticchain-agent.service — systemd unit
- poc/SETUP.md · poc/WIRING.md — full setup + wiring
- docs/ORDER_LIST.md — orderable BOM ×2 + tools
- docs/BUILD_GUIDE.md — 16-week build program
- firmware-hal · mesh-net · compositor · simulator/ — Rust workspace (46 tests)

SUPPORTING DOCS

- docs/HARDWARE.md — fleet overview & design DNA
- docs/feasibility.md — market & technical feasibility
- MEGAPLAN.md — the ignition path (mesh → swarm → sats → launch)

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